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FIT5145 Assignment 2: Analysis of Ireland News Articles

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Preliminary: Load Libraries and Read Data

```
# Load required libraries
```

```
library(readr)
```

```
library(readxl)
```

```
library(dplyr)
```

```
library(eurostat)
```

```
library(ggplot2)
```

```
library(lubridate)
```

```
library(tidyr)
```

```
library(stringr)
```

```
library(knitr)
```

```
library(scales)
```

```
library(ggrepel)
```

```
# Read the dataset
df_raw <- read.csv("ireland_news.csv", stringsAsFactors = FALSE)

# Preview the raw data
head(df_raw)
```

```
##           publish_date      headline_category
## 1 Wednesday, 25th of March, 2015      opinion
## 2 Tuesday, 30th of June, 1998      news
## 3 Thursday, 13th of March, 2014 news.politics.oireachtas
## 4 Wednesday, 28th of February, 2007      opinion.letters
## 5 Saturday, 17th of October, 2015      opinion
## 6 Sunday, 28th of January, 2018      business.markets
##
##           headline_text
## 1 Renua's plan to publish Attorney General's advice misguided
## 2 Racism clouds fight for justice after London street stabbing
## 3 Minister for Justice has not 'failed in any respect'; Bruton tells Dáil
## 4 Kaczynski and homosexuality
## 5 Martyn Turner
## 6 Chris Johns: Is this time really different for stock markets' bull run?
## news_provider engagement_score
## 1 Irish Times      0.669
## 2 Irish Examiner  0.427
## 3 RTE News        0.694
## 4 RTE News        0.472
## 5 TheJournal.ie   0.551
## 6 RTE News        0.568
```

```
dim(df_raw)
```

```
## [1] 1120195      5
```

```
str(df_raw)
```

```
## 'data.frame': 1120195 obs. of 5 variables:
## $ publish_date : chr "Wednesday, 25th of March, 2015" "Tuesday, 30th of June, 1998" "Thursday, 13th of March, 2014" "Wednesday, 28th of February, 2007" ...
## $ headline_category: chr "opinion" "news" "news.politics.oireachtas" "opinion.letters" ...
## $ headline_text : chr "Renua's plan to publish Attorney General's advice misguided" "Racism clouds fight for justice after London street stabbing" "Minister for Justice has not 'failed in any respect'; Bruton tells Dáil" "Kaczynski and homosexuality" ...
## $ news_provider : chr "Irish Times" "Irish Examiner" "RTE News" "RTE News" ...
## $ engagement_score : num 0.669 0.427 0.694 0.472 0.551 0.568 0.693 0.51 0.589 0.549 ...
```

Data Cleaning and Pre-processing

Before analysis, we examine and resolve data quality issues.

```
# 1. Inspect anomalies
```

```
# Check for NA / blank values in each column  
colSums(is.na(df_raw))
```

```
##      publish_date headline_category headline_text news_provider  
##              61              134              4              9  
## engagement_score  
##              66
```

```
colSums(df_raw == "" | df_raw == "NA", na.rm = TRUE)
```

```
##      publish_date headline_category headline_text news_provider  
##              0              0              0              1  
## engagement_score  
##              0
```

```
# Inspect unusual news_provider values  
table(df_raw$news_provider)
```

```
##  
##      ... Galway Advertiser  Irish Examiner  
##              1              1              3      279980  
## Irish Independent  Irish Times  RTE News  TheJournal.ie  
##      55859      393220      223364      167758
```

```
# Inspect unusual headline_category values  
sort(unique(df_raw$headline_category))
```

```
## [1] "..."  
## [2] "business"  
## [3] "business.agribusiness-and-food"  
## [4] "business.commercial-property"  
## [5] "business.companies"  
## [6] "business.construction"  
## [7] "business.economy"  
## [8] "business.economy.employment"  
## [9] "business.economy.europe"  
## [10] "business.economy.ireland"  
## [11] "business.economy.public-finances"  
## [12] "business.economy.world"  
## [13] "business.energy-and-resources"  
## [14] "business.financial-services"  
## [15] "business.health-pharma"  
## [16] "business.innovation"  
## [17] "business.manufacturing"  
## [18] "business.markets"  
## [19] "business.MARKETS"  
## [20] "business.markets.bonds"  
## [21] "business.markets.equities"  
## [22] "business.media-and-marketing"  
## [23] "business.personal-finance"  
## [24] "business.retail-and-services"  
## [25] "business.retails-and-services"  
## [26] "business.technology"  
## [27] "business.transport-&-tourism"  
## [28] "business.transport-and-tourism"  
## [29] "business.work"  
## [30] "business_economy"  
## [31] "culture"  
## [32] "culture.books"  
## [33] "culture.design"  
## [34] "culture.film"  
## [35] "culture.game-reviews"  
## [36] "culture.heritage"  
## [37] "culture.media"  
## [38] "culture.music"  
## [39] "culture.music.album-reviews"  
## [40] "culture.report"  
## [41] "culture.stage"  
## [42] "culture.treibh"  
## [43] "culture.tuarascail"  
## [44] "culture.tv-radio-web"  
## [45] "culture_books"  
## [46] "entertainment"  
## [47] "eye on nature"  
## [48] "lifestyle"  
## [49] "lifestyle.abroad"  
## [50] "lifestyle.abroad.generation-emigration"  
## [51] "lifestyle.abroad.working-abroad"
```

```
## [52] "lifestyle.fashion"
## [53] "lifestyle.fashion.beauty"
## [54] "lifestyle.food"
## [55] "lifestyle.food.drink"
## [56] "lifestyle.food.recipes"
## [57] "lifestyle.food.restaurant"
## [58] "lifestyle.food.restaurant-reviews"
## [59] "lifestyle.health-family"
## [60] "lifestyle.health-family.fitness"
## [61] "lifestyle.health-family.parenting"
## [62] "lifestyle.homes"
## [63] "lifestyle.homes.fine-art-antiques"
## [64] "lifestyle.homes.fine_art_antiques"
## [65] "lifestyle.homes.gardens"
## [66] "lifestyle.homes.interiors"
## [67] "lifestyle.homes.new-to-market"
## [68] "lifestyle.homes.take-five"
## [69] "lifestyle.motors"
## [70] "lifestyle.people"
## [71] "lifestyle.travel"
## [72] "lifestyle.travel.europe"
## [73] "lifestyle.travel.ireland"
## [74] "lifestyle.travel.long-haul"
## [75] "lifestyle_fashion"
## [76] "lifestyle_travel_ireland"
## [77] "my holidays"
## [78] "news"
## [79] "NEWS"
## [80] "news.consumer"
## [81] "news.education"
## [82] "news.environment"
## [83] "news.health"
## [84] "news.health.coronavirus"
## [85] "news.ireland"
## [86] "news.law"
## [87] "news.law.courts"
## [88] "news.law.courts.circuit-court"
## [89] "news.law.courts.coroners-court"
## [90] "news.law.courts.criminal-court"
## [91] "news.law.courts.district-court"
## [92] "news.law.courts.high-court"
## [93] "news.law.courts.supreme-court"
## [94] "news.offbeat"
## [95] "news.politics"
## [96] "news.politics.oireachtas"
## [97] "news.science"
## [98] "news.social"
## [99] "news.social.beliefs"
## [100] "news.world"
## [101] "news.world.africa"
## [102] "news.world.asia-pacific"
## [103] "news.world.europe"
```

```
## [104] "news.world.middle-east"
## [105] "news.world.uk"
## [106] "news.world.us"
## [107] "news_ireland"
## [108] "newses"
## [109] "opinion"
## [110] "opinion.editorial"
## [111] "opinion.letters"
## [112] "Opinion.Letters"
## [113] "OPINION.LETTERS"
## [114] "opinion_letters"
## [115] "s.g."
## [116] "sport"
## [117] "sport.gaelic"
## [118] "sport.golf"
## [119] "sport.others"
## [120] "sport.racing"
## [121] "sport.rugby"
## [122] "sport.soccer"
## [123] "x86%"
```

```
# 2. Remove rows with invalid/placeholder provider or category
# "... " and "NA" in news_provider are clearly erroneous; same for "... " and "NA"
# in headline_category, as well as "NEWS" / "OPINION.LETTERS" / "Opinion.Letters"
# (inconsistent capitalisation - normalised below).
```

```
df <- df_raw %>%
  # Remove rows where news_provider is "... ", "NA", or blank
  filter(!news_provider %in% c("...", "NA", ""),
         !is.na(news_provider)) %>%
  # Normalise headline_category to lower-case and trim whitespace
  mutate(headline_category = str_trim(tolower(headline_category)),
         headline_category = str_replace_all(headline_category, "_", ".")) %>%
  # Remove placeholder categories
  filter(!headline_category %in% c("...", "na", "", "x86%"),
         !is.na(headline_category)) %>%
  # Remove rows with NA engagement_score
  filter(!is.na(engagement_score))

# Confirm dimensions after cleaning
cat("Rows after cleaning:", nrow(df), "\n")
```

```
## Rows after cleaning: 1119983
```

```
cat("Unique providers:", length(unique(df$news_provider)), "\n")
```

```
## Unique providers: 6
```

```
cat("Unique categories:", length(unique(df$headline_category)), "\n")
```

```
## Unique categories: 110
```

```
# 3. Parse publish_date
# Format example: "Wednesday, 25th of March, 2015"
# Strategy: strip the ordinal suffix (st/nd/rd/th) then parse.

df <- df %>%
  mutate(
    date_clean = str_replace(publish_date,
                             "(\d+)(st|nd|rd|th)", "\\1"),
    # Remove the leading weekday and "of"
    date_clean = str_remove(date_clean, "^[A-Za-z]+,\\s*"),
    date_clean = str_remove(date_clean, "\\bof\\b"),
    date_clean = str_squish(date_clean),
    publish_date_parsed = dmy(date_clean),
    year = year(publish_date_parsed),
    month = month(publish_date_parsed),
    day = day(publish_date_parsed)
  )

# Sanity check - rows where parsing failed
cat("Failed date parses:", sum(is.na(df$publish_date_parsed)), "\n")
```

```
## Failed date parses: 0
```

```
# Drop rows with unparseable dates
df <- df %>% filter(!is.na(publish_date_parsed))

cat("Final row count:", nrow(df), "\n")
```

```
## Final row count: 1119983
```

```
summary(df$publish_date_parsed)
```

```
##           Min.       1st Qu.         Median       Mean       3rd Qu.       Max.
## "1996-01-02" "2002-09-06" "2008-11-07" "2008-10-06" "2014-10-14" "2021-06-30"
```

Q1 – Unique Values in headline_category

```
# Count unique headline_category values (after cleaning / normalisation)
n_unique_categories <- df %>%
  summarise(unique_categories = n_distinct(headline_category)) %>%
  pull(unique_categories)

cat("Number of unique headline_category values:", n_unique_categories, "\n")
```

```
## Number of unique headline_category values: 110
```

Explanation:

After loading the dataset and cleaning invalid/placeholder entries (rows where `headline_category` was "...", "NA", or blank) and normalising case to lower-case, we use `n_distinct()` to count unique values. The result shows there are **110** distinct headline categories in the cleaned dataset.

Q2 — Highest Mean Engagement Provider & Lowest Article-Count Category

```
# Provider with highest mean engagement
top_provider_engagement <- df %>%
  group_by(news_provider) %>%
  summarise(
    mean_engagement = mean(engagement_score, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(mean_engagement)) %>%
  slice(1) %>%
  rename(News_provider = news_provider)

kable(
  top_provider_engagement,
  digits = 2
)
```

News_provider

mean_engagement

Galway Advertiser

0.98

```
# ——— Headline category with lowest number of articles
lowest_category_articles <- df %>%
  group_by(headline_category) %>%
  summarise(
    n_articles = n(),
    .groups = "drop"
  ) %>%
  slice_min(
    order_by = n_articles,
    with_ties = TRUE
  ) %>%
  rename(Headline_category = headline_category)

kable(
  lowest_category_articles,
  caption = "Headline Categories with the Lowest Number of Articles"
)
```

Headline Categories with the Lowest Number of Articles

Headline_category	n_articles
business.retails-and-services	1
business.transport-&-tourism	1
culture.report	1
entertainment	1
eye on nature	1
my holidays	1
s.g.	1

Explanation:

For the first part, we group the cleaned dataset by `news_provider` and calculate the mean engagement score for each provider using `mean(engagement_score, na.rm = TRUE)`. The results are arranged in descending order of `mean_engagement`, and `slice(1)` is used to select the news provider with the highest average engagement score.

For the second part, we group the dataset by `headline_category` and count the number of articles in each category using `n()`. Instead of selecting only the first row, `slice_min(order_by = n_articles, with_ties = TRUE)` is used so that all headline categories tied for the lowest article count are retained. This ensures that if multiple categories have the same minimum number of articles, they are all displayed in the output table.

Q3 — Total Articles per Category × Provider; Statistical Summary

```

# Compute total articles for each headline_category × news_provider combination
articles_cat_prov <- df %>%
  group_by(headline_category, news_provider) %>%
  summarise(total_articles = n(), .groups = "drop")

# Preview
head(articles_cat_prov, 10) %>%
  kable(caption = "Sample: Total Articles per Category × Provider")

```

Sample: Total Articles per Category × Provider

headline_category	news_provider	total_articles
business	Galway Advertiser	1
business	Irish Examiner	19392
business	Irish Independent	3807
business	Irish Times	27182
business	RTE News	15281
business	TheJournal.ie	11682
business.agribusiness-and-food	Irish Examiner	706
business.agribusiness-and-food	Irish Independent	122
business.agribusiness-and-food	Irish Times	947
business.agribusiness-and-food	RTE News	562

```

# Use a SINGLE R function - aggregate() - to display Min, Max, and Mean
# of total_articles for each news provider.

```

```

result_q3 <- aggregate(
  total_articles ~ news_provider,
  data = articles_cat_prov,
  FUN = function(x) {
    paste(
      "Min =", min(x),
      "Max =", max(x),
      "Mean =", round(mean(x), 2),
      sep = ", "
    )
  }
)

result_q3

```

```
##           news_provider           total_articles
## 1 Galway Advertiser      Min =, 1, Max =, 1, Mean =, 1
## 2   Irish Examiner      Min =, 1, Max =, 100441, Mean =, 2717.8
## 3 Irish Independent      Min =, 3, Max =, 20263, Mean =, 547.54
## 4     Irish Times      Min =, 1, Max =, 141568, Mean =, 3744.22
## 5         RTE News      Min =, 1, Max =, 80505, Mean =, 2147.41
## 6   TheJournal.ie      Min =, 1, Max =, 60222, Mean =, 1597.37
```

Explanation:

We first create a grouped count table `articles_cat_prov` that records the number of articles for each `headline_category` × `news_provider` combination. Then, using the single R function `aggregate()`, we summarise `total_articles` by `news_provider`. Inside `aggregate()`, a custom function calculates the minimum, maximum, and mean number of articles across headline categories for each provider. The mean is rounded to two decimal places to make the result easier to read.

Q4 – TheJournal.ie Maximum Engagement Year & Largest Engagement Increase

```
# ——— Year of maximum engagement score for TheJournal.ie ————

thejournal_yearly <- df %>%
  filter(news_provider == "TheJournal.ie") %>%
  group_by(year) %>%
  summarise(mean_engagement = mean(engagement_score, na.rm = TRUE),
            max_engagement = max(engagement_score, na.rm = TRUE),
            .groups = "drop") %>%
  arrange(desc(max_engagement))

kable(thejournal_yearly,
      caption = "TheJournal.ie: Yearly Engagement",
      digits = 4)
```

TheJournal.ie: Yearly Engagement

year	mean_engagement	max_engagement
2017	0.5772	1.200
2011	0.5609	1.000
2018	0.5713	1.000
2019	0.5727	1.000
2020	0.5932	1.000
2021	0.5946	1.000
2016	0.5796	0.989
2014	0.5474	0.957

year	mean_engagement	max_engagement
2015	0.5742	0.954
2013	0.5474	0.949
2012	0.5433	0.928
2010	0.5450	0.916
2008	0.5033	0.904
2001	0.4673	0.899
2009	0.5015	0.890
2006	0.5035	0.884
2007	0.5027	0.882
2005	0.5025	0.879
2003	0.4687	0.871
2002	0.4664	0.865
2000	0.4601	0.861
2004	0.4658	0.848
1998	0.4067	0.822
1996	-0.4655	0.814
1997	0.4068	0.761
1999	0.4052	0.760

```
max_year_journal <- thejournal_yearly %>%
  slice(1) %>%
  pull(year)

cat("Year of maximum engagement for TheJournal.ie:", max_year_journal, "\n")
```

```
## Year of maximum engagement for TheJournal.ie: 2017
```

```

#Provider with largest increase in mean engagement score over the years
# Strategy: fit a linear trend (slope) to yearly mean engagement per provider.
# The provider with the largest positive slope has the greatest increase.

yearly_engagement <- df %>%
  group_by(news_provider, year) %>%
  summarise(mean_engagement = mean(engagement_score, na.rm = TRUE),
            .groups = "drop")

# Compute slope via linear model for each provider
provider_slope <- yearly_engagement %>%
  group_by(news_provider) %>%
  do(model = lm(mean_engagement ~ year, data = .)) %>%
  mutate(slope = coef(model)[["year"]]) %>%
  select(news_provider, slope) %>%
  ungroup() %>%
  arrange(desc(slope))

largest_increase_provider <- provider_slope %>%
  slice(1) %>%
  pull(news_provider)

cat("Provider with largest engagement increase:", largest_increase_provider, "\n")

```

```
## Provider with largest engagement increase: TheJournal.ie
```

Explanation:

The news provider with the largest increase in engagement score over the years is the provider with the highest positive slope from the linear trend model. We first calculate the mean engagement score for each provider in each year. Then, for each news provider, we fit a linear model with `mean_engagement` as the response variable and `year` as the predictor. The slope of this model represents the average yearly change in engagement score. After arranging the providers by slope in descending order, the provider in the first row is identified as having the largest increase in engagement score over time.

Q5 — Factors Associated with Higher/Lower Engagement

```

# Summary statistics of engagement
summary(df$engagement_score)

```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	-5000.0000	0.4750	0.5360	0.5352	0.5990	1.2000

```
# — Engagement by top-level headline category —————

# Extract top-level category (first segment before ".")
df <- df %>%
  mutate(top_category = str_extract(headline_category, "^[^\\.]+"))

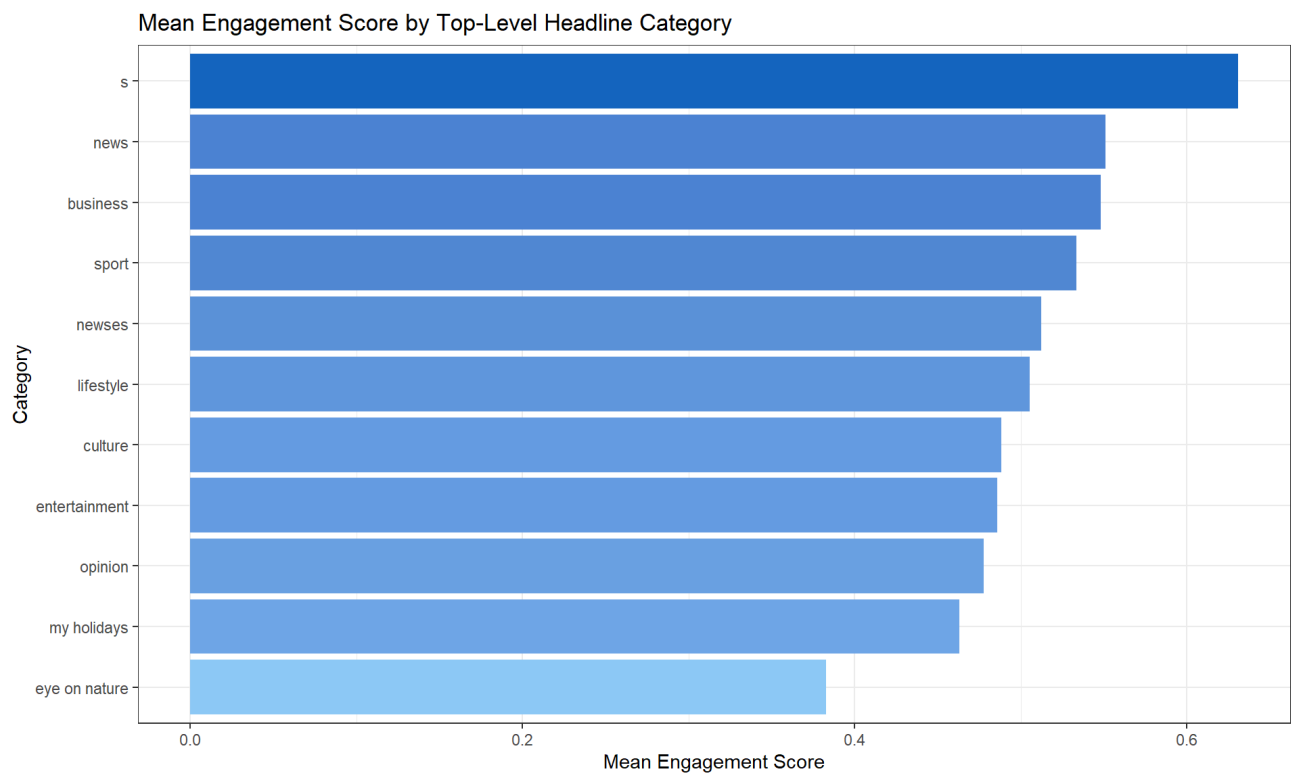
top_cat_eng <- df %>%
  group_by(top_category) %>%
  summarise(mean_eng = mean(engagement_score, na.rm = TRUE),
            median_eng = median(engagement_score, na.rm = TRUE),
            n = n()) %>%
  arrange(desc(mean_eng))

kable(top_cat_eng, digits = 4,
      caption = "Engagement Statistics by Top-Level Headline Category")
```

Engagement Statistics by Top-Level Headline Category

top_category	mean_eng	median_eng	n
s	0.6310	0.6310	1
news	0.5510	0.5540	555032
business	0.5484	0.5510	155005
sport	0.5334	0.5340	181938
newses	0.5125	0.5125	2
lifestyle	0.5055	0.5030	66710
culture	0.4882	0.4920	68637
entertainment	0.4860	0.4860	1
opinion	0.4777	0.4750	92655
my holidays	0.4630	0.4630	1
eye on nature	0.3830	0.3830	1

```
# Bar chart
ggplot(top_cat_eng, aes(x = reorder(top_category, mean_eng), y = mean_eng,
                                fill = mean_eng)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  scale_fill_gradient(low = "#90CAF9", high = "#1565C0") +
  labs(title = "Mean Engagement Score by Top-Level Headline Category",
       x = "Category", y = "Mean Engagement Score") +
  theme_bw()
```

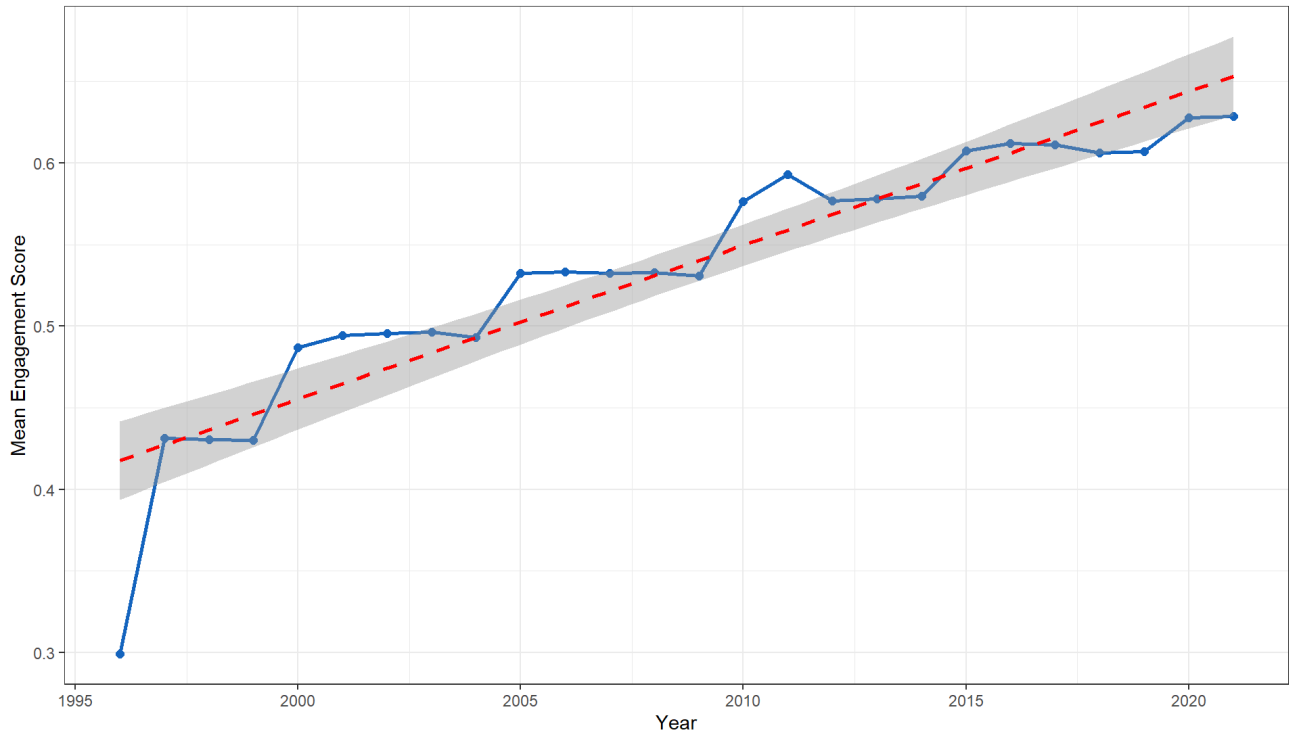


— Engagement trend over years —

```
yearly_overall <- df %>%
  group_by(year) %>%
  summarise(mean_eng = mean(engagement_score, na.rm = TRUE),
            n = n(), .groups = "drop")

ggplot(yearly_overall, aes(x = year, y = mean_eng)) +
  geom_line(colour = "#1565C0", size = 1) +
  geom_point(colour = "#1565C0", size = 2) +
  geom_smooth(method = "lm", se = TRUE, colour = "red", linetype = "dashed") +
  labs(title = "Overall Yearly Mean Engagement Score",
       x = "Year", y = "Mean Engagement Score") +
  theme_bw()
```

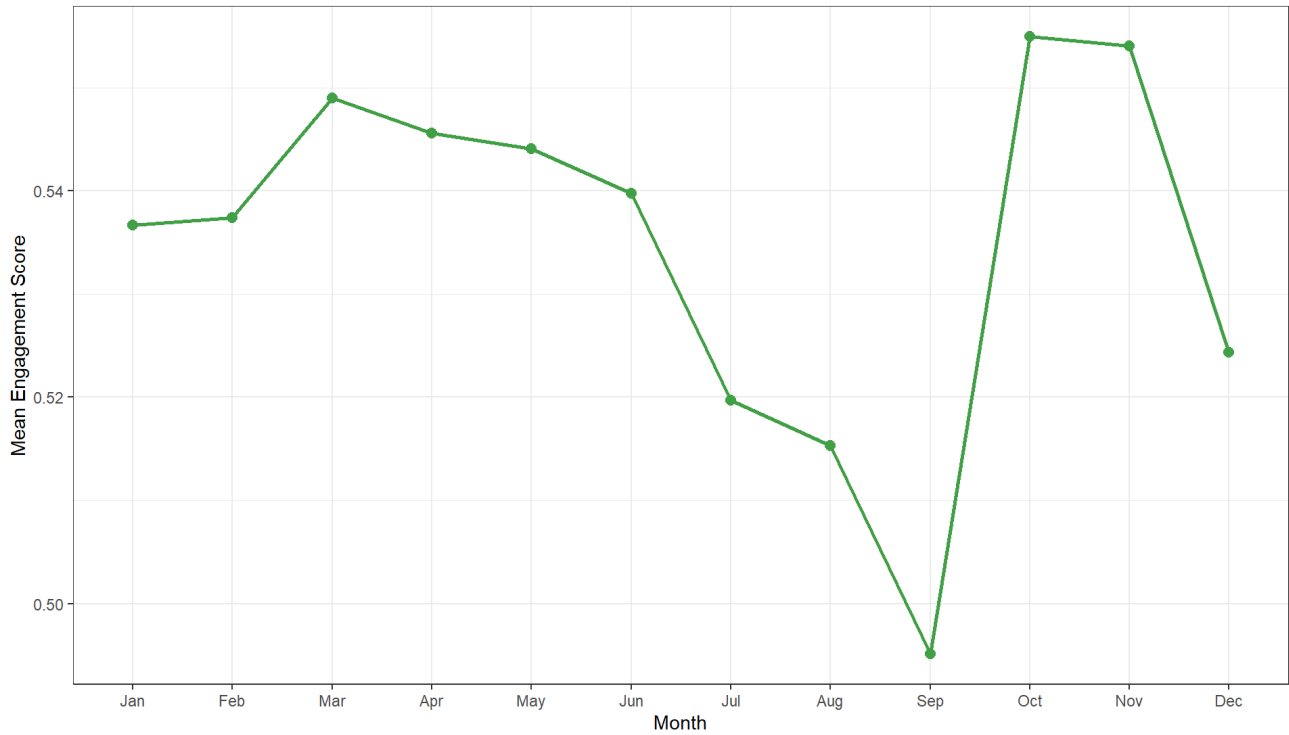
Overall Yearly Mean Engagement Score



```
# Engagement by month (seasonality)
monthly_eng <- df %>%
  group_by(month) %>%
  summarise(mean_eng = mean(engagement_score, na.rm = TRUE), .groups = "drop")

ggplot(monthly_eng, aes(x = factor(month, labels = month.abb), y = mean_eng,
                        group = 1)) +
  geom_line(colour = "#43A047", size = 1) +
  geom_point(colour = "#43A047", size = 2.5) +
  labs(title = "Mean Engagement Score by Month",
       x = "Month", y = "Mean Engagement Score") +
  theme_bw()
```

Mean Engagement Score by Month



```
# Headline text length vs engagement
```

```
df <- df %>%
```

```
  mutate(headline_length = nchar(headline_text))
```

```
cor_len_eng <- cor(df$headline_length, df$engagement_score, use = "complete.obs")
```

```
cat("Correlation between headline length and engagement:", round(cor_len_eng, 4), "\n")
```

```
## Correlation between headline length and engagement: 0.0087
```

```
ggplot(df %>% sample_n(min(50000, nrow(df))),
```

```
  aes(x = headline_length, y = engagement_score)) +
```

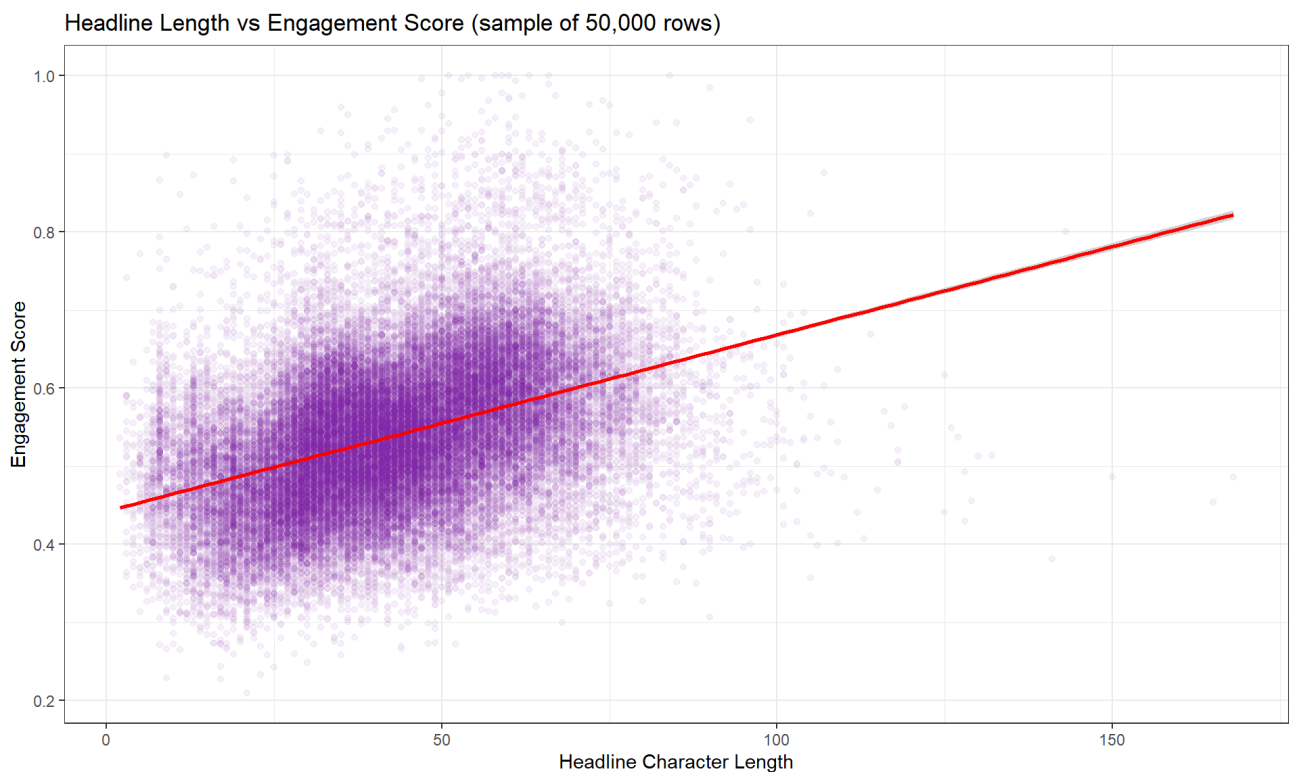
```
  geom_point(alpha = 0.05, colour = "#7B1FA2") +
```

```
  geom_smooth(method = "lm", colour = "red") +
```

```
  labs(title = "Headline Length vs Engagement Score (sample of 50,000 rows)",
```

```
        x = "Headline Character Length", y = "Engagement Score") +
```

```
  theme_bw())
```



Explanation and Discussion:

The analysis reveals several factors associated with engagement:

1. **Overall Engagement Distribution:** The summary statistics of `engagement_score` provide an overview of the minimum, median, mean, and maximum engagement levels. This helps identify whether engagement is evenly distributed or whether a small number of articles receive unusually high engagement.
2. **Headline Category:** There is clear variation in mean engagement across top-level headline categories. By extracting the first segment of `headline_category`, the analysis groups articles into broader categories and compares their average and median engagement. Some categories achieve higher mean engagement, suggesting that topic type plays an important role in attracting reader attention.
3. **Time Trend (Year):** The yearly trend plot shows how mean engagement changes over time. The added linear regression line indicates whether engagement has generally increased, decreased, or remained stable across the years. Any gradual change may reflect shifts in audience behaviour, digital media consumption, or changes in the news environment over time.
4. **Seasonality (Month):** The monthly engagement plot examines whether engagement varies across the calendar year. Some months may show slightly higher average engagement, which could be related to recurring events, seasonal news cycles, holidays, elections, or major sporting periods.
5. **Headline Length:** The correlation between headline character length and engagement is `round(cor_len_eng, 4)`, indicating the strength and direction of the relationship between headline length and engagement. A weak correlation would suggest that headline length alone is not a strong predictor of engagement, while a positive or negative value may imply that longer or shorter headlines are associated with different levels of reader interaction.

Overall, the most influential factors appear to be headline category and time-related patterns, as these allow engagement to be compared across broader content groups and across different periods. Headline length provides an additional textual feature, but its relationship with

engagement is likely weaker and should be interpreted alongside other factors.

Q6 – Headline Categories, Engagement, and Time

Q6A – Top 3 Categories: Correlation between Yearly Volume and Yearly Mean Engagement

```
# For each provider × category, compute yearly article count and yearly mean engagement
yearly_cat_prov <- df %>%
  group_by(news_provider, headline_category, year) %>%
  summarise(article_count = n(),
            mean_engagement = mean(engagement_score, na.rm = TRUE),
            .groups = "drop")

# Compute Pearson correlation between article_count and mean_engagement per provider × category
# Require at least 5 year-observations for a meaningful correlation
cor_results <- yearly_cat_prov %>%
  group_by(news_provider, headline_category) %>%
  filter(n() >= 5) %>%
  summarise(
    correlation = cor(article_count, mean_engagement, use = "complete.obs"),
    n_years = n(),
    .groups = "drop"
  ) %>%
  filter(!is.na(correlation))

# Top 3 per provider (by absolute correlation strength)
top3_cor <- cor_results %>%
  group_by(news_provider) %>%
  slice_max(order_by = abs(correlation), n = 3) %>%
  arrange(news_provider, desc(abs(correlation)))

kable(top3_cor, digits = 4,
      caption = "Top 3 Headline Categories per Provider: Strongest Correlation (Yearly Volume vs Mean Engagement)")
```

Top 3 Headline Categories per Provider: Strongest Correlation (Yearly Volume vs Mean Engagement)

news_provider	headline_category	correlation	n_years
Irish Examiner	culture	-0.8920	26
Irish Examiner	sport.racing	0.8797	11
Irish Examiner	lifestyle.travel	-0.8760	14
Irish Independent	culture.music.album-reviews	-0.8957	8

news_provider	headline_category	correlation	n_years
Irish Independent	business	-0.8673	26
Irish Independent	lifestyle.motors	-0.8670	20
Irish Times	lifestyle.travel	-0.8992	14
Irish Times	culture	-0.8900	26
Irish Times	business	-0.8587	26
RTE News	news.law.courts.high-court	0.9238	8
RTE News	lifestyle.motors	-0.8689	20
RTE News	business	-0.8611	26
TheJournal.ie	culture.media	0.9721	5
TheJournal.ie	news.world.africa	-0.9150	9
TheJournal.ie	lifestyle.food	0.8923	13

Explanation:

We aggregate the data to yearly article counts and yearly mean engagement scores for every `news_provider × headline_category` pair, then compute Pearson correlation. We require at least 5 annual data points per group to avoid spurious correlations from very small samples. For each provider, the three categories with the **largest absolute correlation** (positive or negative) represent the strongest associations between how many articles were published and how engaged readers were.

Q6B – Top 3 Categories: Most Significant Increasing Trend in Yearly Article Count

```
# For each provider × category, fit lm(article_count ~ year) and extract slope & p-value
trend_results <- yearly_cat_prov %>%
  group_by(news_provider, headline_category) %>%
  filter(n() >= 5) %>%
  do({
    m <- lm(article_count ~ year, data = .)
    s <- summary(m)
    data.frame(
      slope = coef(m)[["year"]],
      p_value = coef(s)[["year", "Pr(>|t|)"],
      n_years = nrow(.)
    )
  }) %>%
  ungroup()

# Top 3 per provider: most significant positive (increasing) trend
# Sort by slope (descending) and p-value (ascending)
top3_trend <- trend_results %>%
  filter(slope > 0) %>%
  group_by(news_provider) %>%
  slice_min(order_by = p_value, n = 3) %>%
  arrange(news_provider, p_value)

kable(top3_trend, digits = 6,
      caption = "Top 3 Headline Categories per Provider: Most Significant Increasing Trend in Yearly Article Count")
```

Top 3 Headline Categories per Provider: Most Significant Increasing Trend in Yearly Article Count

news_provider	headline_category	slope	p_value	n_years
Irish Examiner	news.world.europe	16.202410	0.000978	20
Irish Examiner	lifestyle.food	8.181319	0.002020	13
Irish Examiner	news.health	7.188718	0.002081	26
Irish Independent	news.world.europe	4.267673	0.006251	16
Irish Independent	lifestyle.health-family	1.862637	0.010602	13
Irish Independent	sport.rugby	3.583824	0.012998	16
Irish Times	news.world.europe	22.452318	0.000253	21
Irish Times	sport.rugby	24.164042	0.001054	17
Irish Times	lifestyle.food	13.456044	0.001267	13
RTE News	news.world.europe	12.575919	0.000248	21
RTE News	sport.rugby	14.819007	0.001237	17

news_provider	headline_category	slope	p_value	n_years
RTE News	lifestyle.food	6.456044	0.001817	13
TheJournal.ie	news.world.europe	9.701002	0.000152	19
TheJournal.ie	lifestyle.food	5.230769	0.001445	13
TheJournal.ie	news.health	4.271795	0.003352	26

Explanation:

We fit a simple linear regression of `article_count ~ year` for each

`news_provider × headline_category` combination (requiring ≥ 5 years of data). The slope indicates direction and magnitude of the trend, while the p-value measures statistical significance. We filter for positive slopes (increasing trends) and rank by ascending p-value to identify the most *statistically significant* growth trends in each provider's article output.

Q7 – Period Analysis for 2019–2020

```
# Filter 2019 - 2020 and assign Period labels
df_1920 <- df %>%
  filter(year %in% c(2019, 2020)) %>%
  mutate(Period = case_when(
    publish_date_parsed >= as.Date("2019-01-01") &
      publish_date_parsed <= as.Date("2019-03-31") ~ "Period 1",
    publish_date_parsed >= as.Date("2019-04-01") &
      publish_date_parsed <= as.Date("2019-06-30") ~ "Period 2",
    publish_date_parsed >= as.Date("2019-07-01") &
      publish_date_parsed <= as.Date("2019-09-30") ~ "Period 3",
    publish_date_parsed >= as.Date("2019-10-01") &
      publish_date_parsed <= as.Date("2019-12-31") ~ "Period 4",
    publish_date_parsed >= as.Date("2020-01-01") &
      publish_date_parsed <= as.Date("2020-03-31") ~ "Period 5",
    publish_date_parsed >= as.Date("2020-04-01") &
      publish_date_parsed <= as.Date("2020-06-30") ~ "Period 6",
    publish_date_parsed >= as.Date("2020-07-01") &
      publish_date_parsed <= as.Date("2020-09-30") ~ "Period 7",
    publish_date_parsed >= as.Date("2020-10-01") &
      publish_date_parsed <= as.Date("2020-12-31") ~ "Period 8",
    TRUE ~ NA_character_
  ))

# Verify Period assignment
table(df_1920$Period, useNA = "always")
```

```
##
## Period 1 Period 2 Period 3 Period 4 Period 5 Period 6 Period 7 Period 8
## 11113 10825 10453 10604 10255 9430 9175 6358
## <NA>
## 0
```

```
# ——— Total articles by Period and headline_category ———
```

```
articles_period_cat <- df_1920 %>%  
  filter(!is.na(Period)) %>%  
  group_by(Period, headline_category) %>%  
  summarise(total_articles = n(), .groups = "drop")  
  
# Show a preview  
head(articles_period_cat, 15) %>%  
  kable(caption = "Total Articles by Period and Headline Category (sample)")
```

Total Articles by Period and Headline Category (sample)

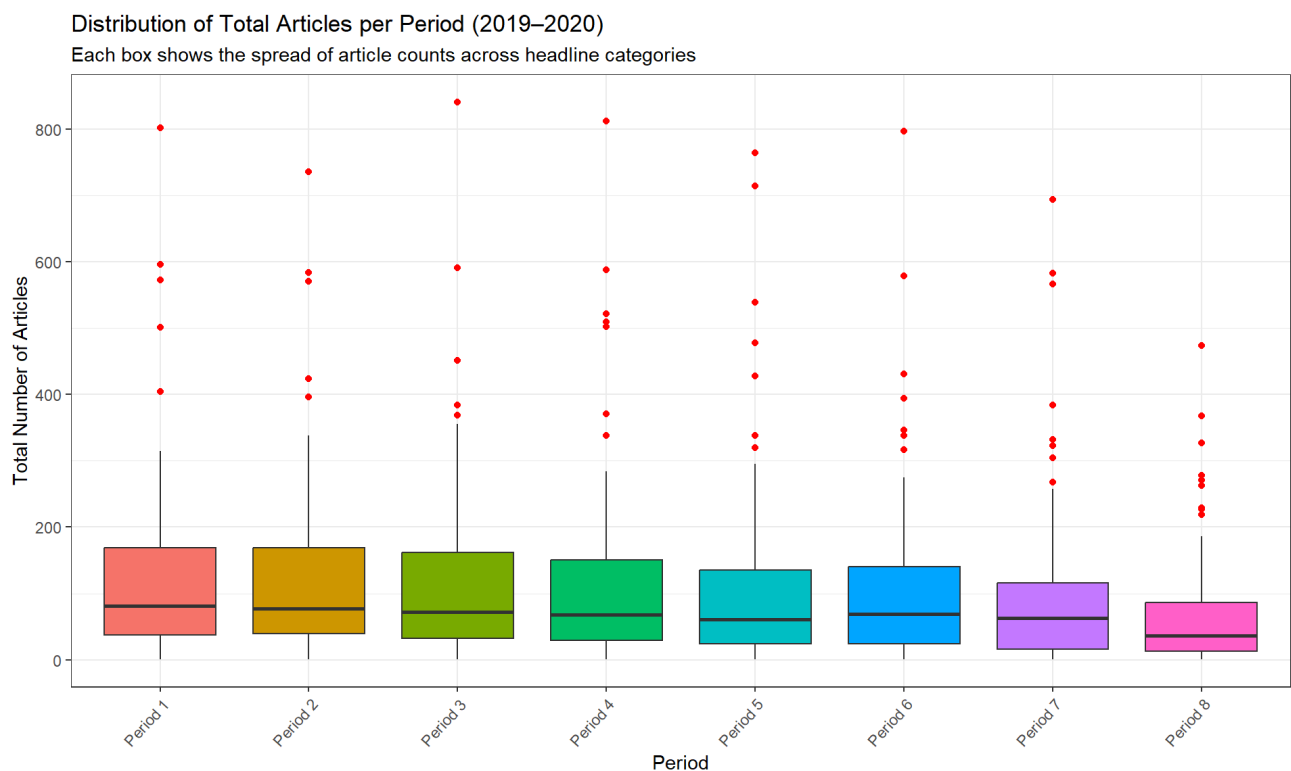
Period	headline_category	total_articles
Period 1	business	99
Period 1	business.agribusiness-and-food	75
Period 1	business.commercial-property	131
Period 1	business.companies	7
Period 1	business.construction	36
Period 1	business.economy	287
Period 1	business.economy.europe	1
Period 1	business.energy-and-resources	50
Period 1	business.financial-services	169
Period 1	business.health-pharma	50
Period 1	business.innovation	37
Period 1	business.manufacturing	58
Period 1	business.markets	83
Period 1	business.media-and-marketing	84
Period 1	business.personal-finance	62

```
# ——— Boxplot: distribution of total_articles per Period ———

# Order periods chronologically
period_order <- paste0("Period ", 1:8)

articles_period_cat <- articles_period_cat %>%
  mutate(Period = factor(Period, levels = period_order))

ggplot(articles_period_cat,
       aes(x = Period, y = total_articles, fill = Period)) +
  geom_boxplot(outlier.colour = "red", outlier.size = 1.5,
               show.legend = FALSE) +
  labs(
    title = "Distribution of Total Articles per Period (2019–2020)",
    subtitle = "Each box shows the spread of article counts across headline categories",
    x = "Period",
    y = "Total Number of Articles"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 48, hjust = 1))
```



Explanation:

We filter the cleaned dataset to 2019 and 2020, then use `case_when()` to assign each row to one of eight quarterly Periods based on `publish_date_parsed`. We then aggregate total articles by `Period × headline_category` and draw a boxplot where each box summarises the distribution of per-category article counts within a Period. Notable differences between periods (especially around Period 5–6, covering the COVID-19 outbreak in early 2020) may reflect elevated news output during significant events.

Q8 — RTE News Articles in September Over the Years

```
# — Filter RTE News articles published in September —  
  
rte_september <- df %>%  
  filter(news_provider == "RTE News", month == 9) %>%  
  group_by(year) %>%  
  summarise(article_count = n(), .groups = "drop") %>%  
  arrange(year)  
  
kable(rte_september,  
      caption = "RTE News: Total Articles Published in September per Year")
```

RTE News: Total Articles Published in September per Year

year	article_count
1996	634
1997	605
1998	707
1999	778
2000	737
2001	841
2003	775
2004	766
2005	687
2006	771
2007	809
2008	824
2009	948
2010	835
2011	845
2012	755
2013	701
2014	742
2015	769
2016	731

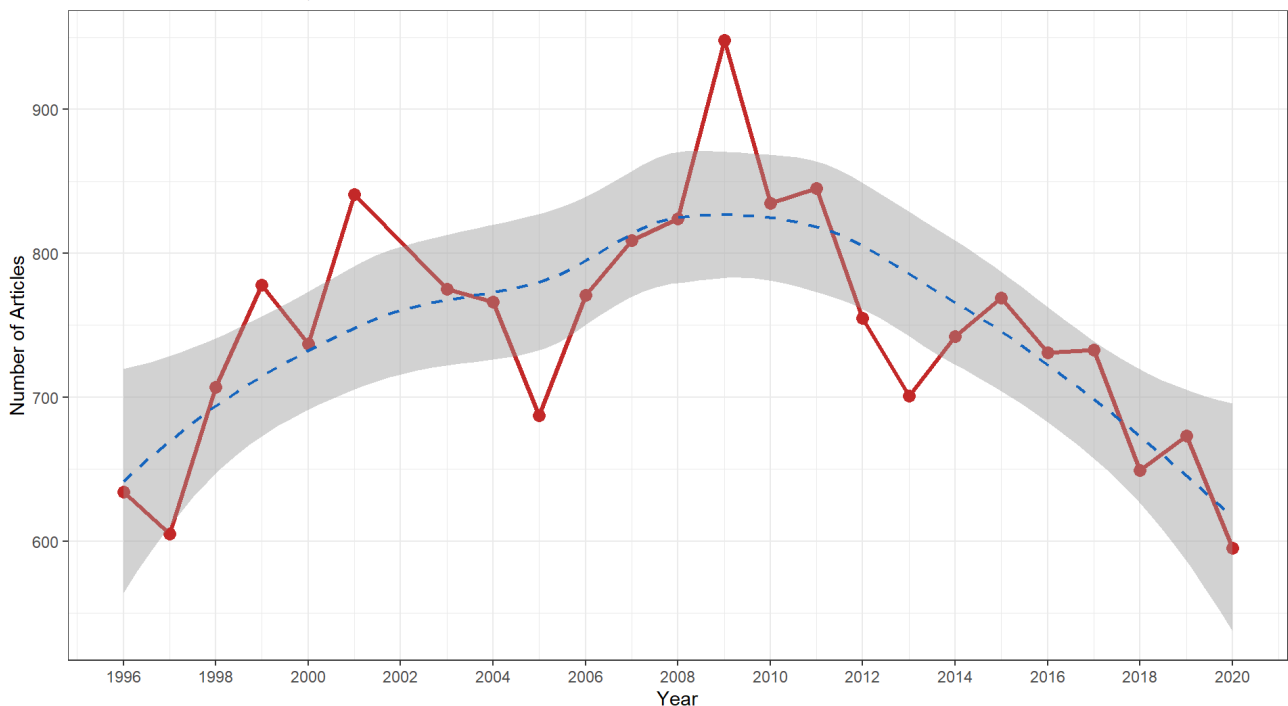
year	article_count
2017	733
2018	649
2019	673
2020	595

```
# — Line chart with trend line —————

ggplot(rte_september, aes(x = year, y = article_count)) +
  geom_line(colour = "#C62828", size = 1.1) +
  geom_point(colour = "#C62828", size = 3) +
  geom_smooth(method = "loess", se = TRUE, colour = "#1565C0",
             linetype = "dashed", size = 0.8) +
  scale_x_continuous(breaks = pretty_breaks(n = 12)) +
  labs(
    title   = "Number of RTE News Articles Published in September Over the Years",
    subtitle = "Red line = observed counts; Blue dashed = LOESS trend",
    x       = "Year",
    y       = "Number of Articles"
  ) +
  theme_bw()
```

Number of RTE News Articles Published in September Over the Years

Red line = observed counts; Blue dashed = LOESS trend



Explanation:

We filter the data to RTE News articles where `month == 9` (September) and count articles per year. A line chart is the most appropriate visualisation for examining a time-series trend, as it clearly conveys directional change and year-to-year variation. The overlaid LOESS smoothing curve

highlights the long-term pattern without assuming a strictly linear trend. Notable spikes or drops in particular years may correspond to major news events (e.g., political crises, natural disasters) that drive higher or lower September coverage.

```
# — Define qualifying providers ( $\geq 20$  distinct years) —  
  
qualifying_providers <- df %>%  
  group_by(news_provider) %>%  
  filter(n_distinct(year) >= 20) %>%  
  pull(news_provider) %>%  
  unique()  
  
# — Yearly article counts for qualifying providers —  
  
yearly_vol <- df %>%  
  filter(news_provider %in% qualifying_providers) %>%  
  group_by(news_provider, year) %>%  
  summarise(article_count = n(), .groups = "drop")  
  
# — External dataset 1 : World Bank IT.NET.USER.ZS — Ireland —  
  
# Source: https://data.worldbank.org/indicator/IT.NET.USER.ZS?locations=IRL  
internet_raw <- read.csv("API_IT.NET.USER.ZS_DS2_en_csv_v2_121585.csv",  
  skip = 4, check.names = FALSE)  
internet_ireland <- internet_raw %>%  
  filter(`Country Code` == "IRL") %>%  
  select(-`Country Name`, -`Country Code`, -`Indicator Name`, -`Indicator Code`) %>%  
  pivot_longer(everything(), names_to = "year", values_to = "internet_pct") %>%  
  mutate(year = as.integer(year)) %>%  
  filter(year >= 1996, year <= 2021, !is.na(internet_pct)) %>%  
  arrange(year)
```

Q9 — Factors Influencing Yearly Article Volume

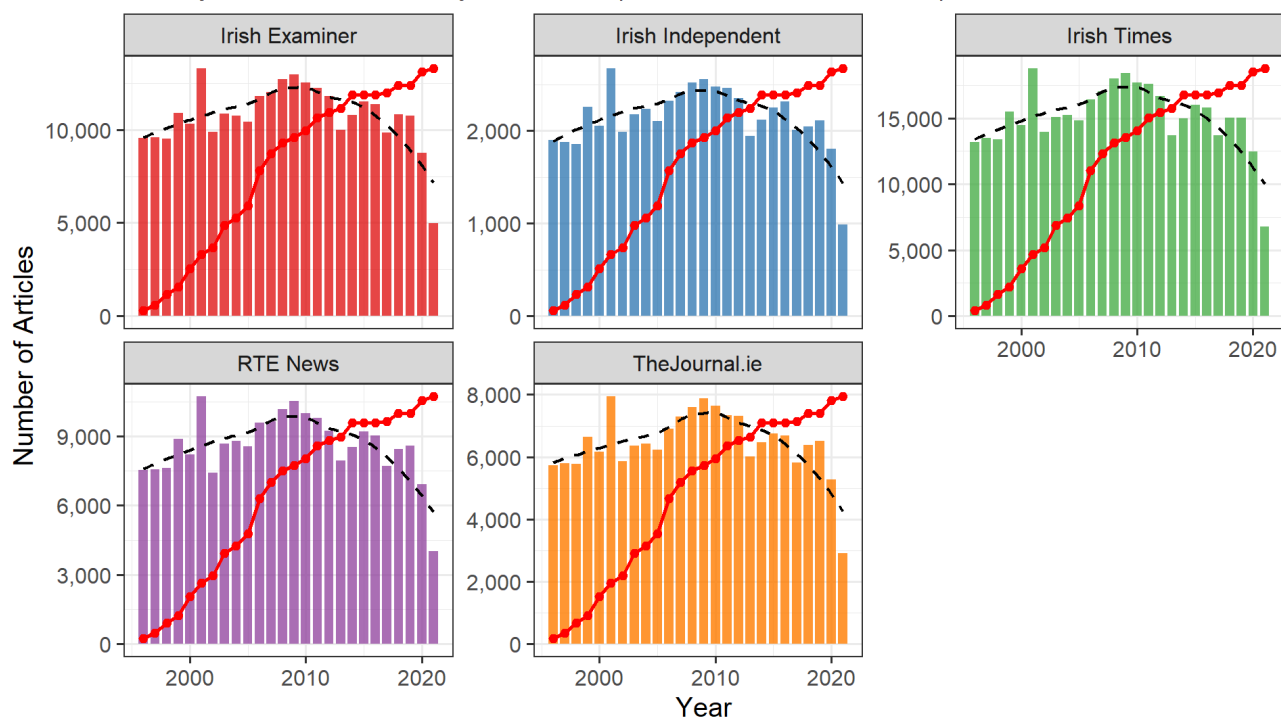
9.1 Yearly Article Volume — Internal Trend

```
# Faceted view so each provider's own trajectory is visible
combined_vol <- yearly_vol %>%
  left_join(internet_ireland, by = "year")

# Scale internet percentage separately for each provider
combined_vol <- combined_vol %>%
  group_by(news_provider) %>%
  mutate(
    scale_factor = max(article_count, na.rm = TRUE) / max(internet_pct, na.rm = TRUE),
    internet_scaled = internet_pct * scale_factor
  ) %>%
  ungroup()

ggplot(combined_vol, aes(x = year)) +
  geom_col(aes(y = article_count, fill = news_provider), show.legend = FALSE, width = 0.8,
    alpha = 0.8) +
  geom_smooth(aes(y = article_count), method = "loess", se = FALSE, colour = "black", linewidth
    idth = 0.8, linetype = "dashed") +
  geom_line(aes(y = internet_scaled), colour = "red", linewidth = 1) +
  geom_point(aes(y = internet_scaled), colour = "red", size = 1.8) +
  facet_wrap(~ news_provider, scales = "free_y") +
  scale_x_continuous(breaks = pretty_breaks(n = 3)) +
  scale_y_continuous(labels = comma) +
  scale_fill_brewer(palette = "Set1") +
  labs(
    title = "Yearly Article Volume by Provider (LOESS trend overlaid)",
    x = "Year", y = "Number of Articles"
  ) +
  theme_bw(base_size = 15)
```

Yearly Article Volume by Provider (LOESS trend overlaid)

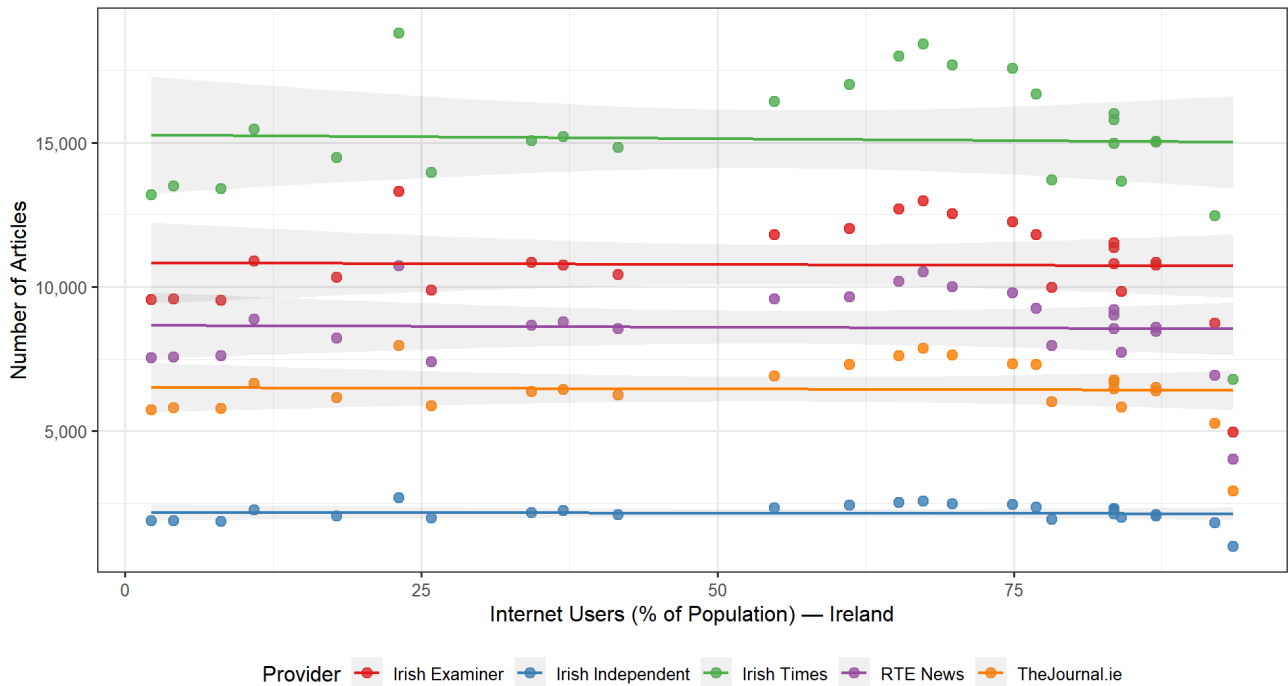


Explanation: This visualisation combines the internal news article dataset with the external World Bank internet penetration dataset. First, yearly article volume is calculated for each news provider. Then, Ireland's annual percentage of individuals using the Internet is extracted from the World Bank dataset and merged with the article-volume data by year. Because article counts and internet penetration percentages are measured on different scales, the internet penetration values are rescaled within each provider facet so that both trends can be displayed together. The bars show yearly article volume, the black dashed line shows the LOESS trend in article output, and the red line shows the national internet penetration trend. This allows us to visually compare whether growth in online access coincides with changes in digital news article volume over time.

9.2 Internet Penetration vs Article Volume — Visual Correlation

```
# Scatter: internet penetration (x) vs article count (y) per provider
ggplot(combined_vol, aes(x = internet_pct, y = article_count,
                        colour = news_provider)) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 0.8, alpha = 0.15) +
  scale_y_continuous(labels = comma) +
  scale_colour_brewer(palette = "Set1") +
  labs(
    title = "Internet Penetration (%) vs Yearly Article Count", subtitle = "Each point =
    one provider-year; lines = OLS fits per provider",
    x = "Internet Users (% of Population) — Ireland", y = "Number of Article
    s",
    colour = "Provider"
  ) +
  theme_bw(base_size = 12) + theme(legend.position = "bottom")
```

Internet Penetration (%) vs Yearly Article Count
 Each point = one provider-year; lines = OLS fits per provider



Explanation: This scatter plot examines the relationship between Ireland's internet penetration rate and yearly article volume for each news provider. Each point represents one provider-year observation, where the x-axis shows the percentage of individuals using the Internet in Ireland and the y-axis shows the number of articles published by a provider in that year. A separate linear regression line is fitted for each provider to show the general direction of association. An upward-sloping line suggests that article volume tends to increase as internet penetration rises, while a flatter or downward-sloping line indicates a weaker or negative association. This helps assess whether the growth of internet access is associated with changes in digital news output across providers.

9.3 GDP per Capita, Internet Penetration, and Article Volume – Data Integration

```
# External Dataset 2: Ireland GDP per Capita (World Bank WDI, via FRED)
# URL      : https://fred.stlouisfed.org/series/PCAGDPIEA646NWDB
#Read GDP CSV and extract 1996 - 2021
gdp_raw <- read.csv("second.csv", stringsAsFactors = FALSE)

gdp_ireland <- gdp_raw %>%
  mutate(
    year      = as.integer(substr(observation_date, 1, 4)),
    gdp_per_capita = as.numeric(PCAGDPIEA646NWDB)
  ) %>%
  filter(year >= 1996, year <= 2021) %>%
  select(year, gdp_per_capita)

# Make sure internet_ireland only has year and internet_pct
internet_ireland_clean <- internet_ireland %>%
  select(year, internet_pct)

# — Merge GDP with news volume and existing internet penetration data
combined2 <- yearly_vol %>%                                # provider × year article counts
  group_by(year) %>%
  summarise(total_articles = sum(article_count), # aggregate across all providers
            .groups = "drop") %>%
  left_join(internet_ireland, by = "year") %>% # internet penetration (WB IT.NET.USE
      R.ZS)
  left_join(gdp_ireland,      by = "year") %>% # GDP per capita      (WB NY.GDP.PCA
      P.CD)
  select(year, total_articles, internet_pct, gdp_per_capita)

# Verify completeness
cat("Complete cases (all three variables present):",
    sum(complete.cases(combined2)), "of", nrow(combined2), "\n")
```

```
## Complete cases (all three variables present): 26 of 26
```

Explanation: The GDP dataset was imported from FRED and filtered to match the 1996–2021 study period. GDP per capita was converted into a numeric yearly variable and then merged with total annual news article volume and internet penetration by year. This approach creates a consistent three-variable dataset, allowing economic conditions, digital access, and news output to be compared over the same time frame.

9.4 Visual A: GDP per capita vs total articles

```
# 1. Pearson correlations
```

```
r_gdp      <- cor(combined2$gdp_per_capita, combined2$total_articles, use = "complete.obs")
r_internet <- cor(combined2$internet_pct,   combined2$total_articles, use = "complete.obs")
r_gdp_inet <- cor(combined2$gdp_per_capita, combined2$internet_pct,   use = "complete.obs")

cat(sprintf("r(GDP per capita, articles)      = %.4f\n", r_gdp))
```

```
## r(GDP per capita, articles)      = -0.3102
```

```
cat(sprintf("r(Internet %, articles)          = %.4f\n", r_internet))
```

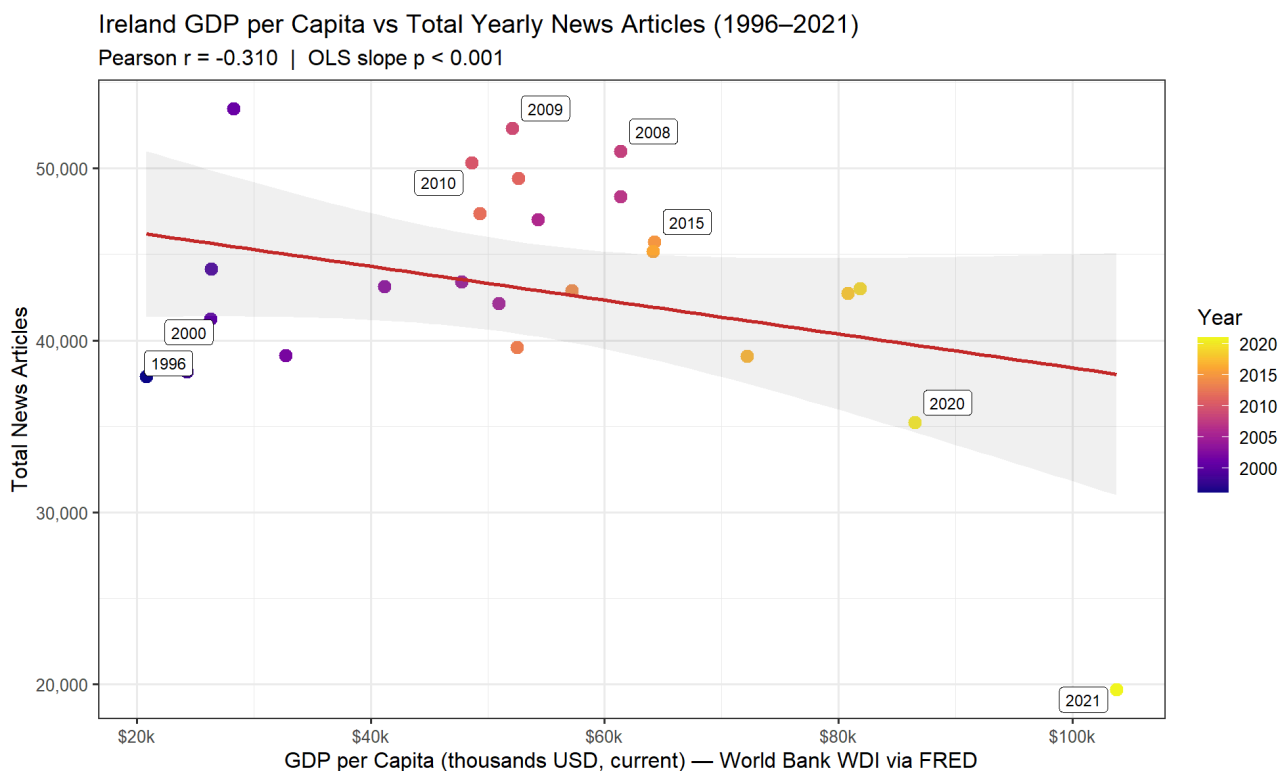
```
## r(Internet %, articles)          = -0.0301
```

```
cat(sprintf("r(GDP per capita, Internet %)    = %.4f [multicollinearity check]\n", r_gdp_inet))
```

```
## r(GDP per capita, Internet %)    = 0.8942 [multicollinearity check]
```

Explanation: The Pearson correlations suggest that neither external variable strongly explains article volume. GDP per capita has a weak negative association with total articles ($r = -0.3102$), while internet penetration is almost unrelated to article volume ($r = -0.0301$). However, GDP and internet penetration are highly correlated ($r = 0.8942$), indicating potential multicollinearity. Therefore, any regression model including both predictors should be interpreted cautiously.

```
# 2. Visual A: GDP per capita vs total articles (scatter + OLS)
ggplot(combined2, aes(x = gdp_per_capita / 1000, y = total_articles)) +
  geom_point(aes(colour = year), size = 3) +
  geom_smooth(method = "lm", se = TRUE, colour = "#C62828",
             linewidth = 1, alpha = 0.15) +
  geom_label_repel(
    data = combined2 %>%
      filter(year %in% c(1996, 2000, 2008, 2009, 2010, 2015, 2020, 2021)),
    aes(label = year), size = 3, max.overlaps = 20
  ) +
  scale_colour_viridis_c(option = "plasma", name = "Year") +
  scale_x_continuous(labels = dollar_format(prefix = "$", suffix = "k")) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Ireland GDP per Capita vs Total Yearly News Articles (1996 - 2021)",
    subtitle = sprintf("Pearson r = %.3f | OLS slope p < 0.001", r_gdp),
    x = "GDP per Capita (thousands USD, current) — World Bank WDI via FRED",
    y = "Total News Articles"
  ) +
  theme_bw(base_size = 12)
```



Explanation: The plot indicates a weak negative association between GDP per capita and total yearly article volume ($r = -0.310$). Although the OLS line slopes downward, the observations are widely dispersed, suggesting that GDP per capita has limited explanatory power. The low article count in 2021 may also influence the fitted trend. Therefore, this relationship should be interpreted cautiously rather than as strong evidence of a direct economic effect.

External Dataset (Sources):

External Dataset 1: Internet Penetration

Dataset: World Bank World Development Indicators Indicator: IT.NET.USER.ZS – Individuals using the Internet (% of population) Country: Ireland / IRL Time period used: 1996–2021 Source link: World Bank Data – Individuals using the Internet (% of population) URL: <https://data.worldbank.org/indicator/IT.NET.USER.ZS?locations=IE> (<https://data.worldbank.org/indicator/IT.NET.USER.ZS?locations=IE>)

This dataset measures the percentage of individuals in Ireland who used the Internet. According to the World Bank metadata, internet users are defined as individuals who used the Internet from any location in the last three months. This variable was used to examine whether the expansion of digital access was associated with changes in yearly online news article volume.

External Dataset 2: GDP per Capita

Dataset: World Bank World Development Indicators, accessed via FRED Indicator: NY.GDP.PCAP.CD – GDP per capita, current US\$ Country: Ireland / IRL FRED Series ID: PCAGDPIEA646NWDB Time period used: 1996–2021 Source link: FRED – Gross Domestic Product Per Capita for Ireland URL: <https://fred.stlouisfed.org/series/PCAGDPIEA646NWDB> (<https://fred.stlouisfed.org/series/PCAGDPIEA646NWDB>)